CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

SOARES DAIRY #4

9207 AMERICAN AVE

HILMAR CA 95324

NOTES:

PREPARED BY:

CARDOSO AG SERVICES
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Joe Soares Dairy #4

Physical address of dairy:

9207 American Ave	Hilmar	Stanislaus	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1930

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X020-X004-XXXX X057-X023-X009-XXXX X057-X023-X017-XXXX

B. OPERATORS

Soares, Joe & Teresa

Operator name: Soares, Joe & Teresa

Telephone no.: Landline: ; Cellular: (209) 652-2990

P.O. Box 189	Delhi	CA	95135
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Soares, Joe & Teresa

Legal owner name: Soares, Joe & Teresa

Telephone no.: Landline: ; Cellular: (209) 652-2990

P.O. Box 189	Delhi	CA	95135
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	1,074	112	0	0
Number under roof	900	0	0	0	0	0
Maximum number	900	0	1,074	112	0	0
Average number	900	0	1,074	112	0	0
Avg live weight (lbs)	1,400	0	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 35,966.83 *tons per reporting period*

Total nitrogen from manure: 417,155.52 *lbs per reporting period*

After ammonia losses (30% loss applied): 292,008.86 *lbs per reporting period*

Total phosphorus from manure: 70,655.07 *lbs per reporting period*

Total potassium from manure: 173,540.90 *lbs per reporting period*

Total salt from manure: 423,765.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 28,260,000 *gallons*

Total nitrogen generated: 119,741.91 *lbs*

Total phosphorus generated: 19,376.56 *lbs*

Total potassium generated: 240,829.44 *lbs*

Total salt generated: 1,329,542.51 *lbs*

	10,260,000	*gallons applied*
+	18,000,000	*gallons exported*
-	0	*gallons imported*
=	28,260,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	Corral solids	3,820.00 *ton*	Dry-weight	56.2		23,300.00	10,800.00	39,100.00		0.00
10/18/2023	Corral solids	4,840.00 *ton*	Dry-weight	46.8		16,800.00	5,800.00	23,900.00		0.00

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
05/08/2023	Process wastewater	5,850,000.00 *gal*	757.00	438.00	0.00	0.90	116.00	1,400.00		7,000
06/20/2023	Process wastewater	5,850,000.00 *gal*	757.00	438.00	0.00	0.90	116.00	1,400.00		7,000
08/02/2023	Process wastewater	6,300,000.00 *gal*	244.00	174.00	0.00	11.80	46.50	609.00		4,180

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	164,485.22	66,008.86	253,920.38	0.00
Process wastewater	87,447.00	13,770.50	168,708.36	903,212.73
Total exports for all materials	251,932.23	79,779.37	422,628.74	903,212.73

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	29	29	2	both	X057-X023-X001-XXXX
Field 2	21	21	2	both	X057-X023-X002-XXXX
Field 3	19	19	2	both	X057-X023-X003-XXXX
Field 4	21	21	2	both	X057-X023-X017-XXXX
Totals for areas that were used for application	90	90	8		
Totals for areas that were not used for application					
Land application area totals	90	90	8		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	404.00 *ton*	Dry-weight		61.3	14,300.00	3,100.00	31,100.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.93	154.19	33.43	335.34	1,142.96

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	802.00 *ton*	Dry-weight		68.9	12,800.00	1,800.00	15,900.00		1.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.66	220.18	30.96	273.50	326.83

Field 2

Field name: Field 2

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	292.00 *ton*	Dry-weight		66.0	16,500.00	4,000.00	34,800.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.90	156.01	37.82	329.04	1,087.35

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	581.00 *ton*	Dry-weight		69.0	13,200.00	2,000.00	13,500.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.67	226.42	34.31	231.57	823.36

Field 3

Field name: Field 3

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	265.00 *ton*	Dry-weight		61.7	16,200.00	4,300.00	37,500.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.95	173.08	45.94	400.64	1,346.14

Field 3

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	526.00 *ton*	Dry-weight		70.9	13,600.00	1,900.00	19,100.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.68	219.13	30.61	307.74	853.95

Field 4

Field name: Field 4

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	292.00 *ton*	Dry-weight		64.6	13,700.00	3,300.00	36,900.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.90	134.87	32.49	363.26	1,210.88

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	581.00 *ton*	Dry-weight		71.0	14,000.00	1,900.00	19,100.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.67	224.65	30.49	306.49	1,026.99

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/11/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	61.65	14.32	75.98	0.00	163.00 *ton*
Application event totals		61.65	14.32	75.98	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.17	9.08	134.33	812.17	630,000.00 *gal*
Canal	Surface water	5.52	0.00	0.00	147.09	2,999,747.00 *gal*
Application event totals		41.69	9.08	134.33	959.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.14	15.02	181.29	906.44	450,000.00 *gal*
Canal	Surface water	7.37	0.00	0.00	196.12	3,999,662.00 *gal*
Application event totals		105.51	15.02	181.29	1,102.56	

Field 1 - 06/20/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.14	15.02	181.29	906.44	450,000.00 *gal*
Canal	Surface water	11.66	0.00	0.00	310.52	6,332,834.00 *gal*
Application event totals		109.80	15.02	181.29	1,216.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.75	0.00	0.00	232.89	4,749,626.00 *gal*
Application event totals		8.75	0.00	0.00	232.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	44.72	8.13	106.46	730.72	607,500.00 *gal*
Canal	Surface water	12.28	0.00	0.00	326.87	6,666,122.00 *gal*
Application event totals		56.99	8.13	106.46	1,057.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	39.75	7.23	94.63	649.53	540,000.00 *gal*
Canal	Surface water	9.82	0.00	0.00	261.49	5,332,919.00 *gal*
Application event totals		49.57	7.23	94.63	911.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.37	8.43	110.40	757.78	630,000.00 *gal*
Canal	Surface water	7.37	0.00	0.00	196.12	3,999,689.00 *gal*
Application event totals		53.74	8.43	110.40	953.90	

Field 1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	13.35	0.00	0.00	355.47	7,249,401.00 *gal*
Application event totals		13.35	0.00	0.00	355.47	

09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.90	0.00	0.00	236.98	4,832,934.00 *gal*
Application event totals		8.90	0.00	0.00	236.98	

Field 2 - 10/11/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	64.77	15.05	79.82	0.00	124.00 *ton*
Application event totals		64.77	15.05	79.82	0.00	

11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.46	9.41	139.13	841.18	472,500.00 *gal*
Canal	Surface water	7.80	0.00	0.00	207.64	3,066,393.00 *gal*
Application event totals		45.26	9.41	139.13	1,048.81	

Field 2 - 10/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.87	14.52	175.25	876.23	315,000.00 *gal*
Canal	Surface water	6.69	0.00	0.00	178.10	2,630,143.00 *gal*
Application event totals		101.56	14.52	175.25	1,054.32	

Field 2 - 06/20/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.87	14.52	175.25	876.23	315,000.00 *gal*
Canal	Surface water	9.30	0.00	0.00	247.72	3,658,342.00 *gal*
Application event totals		104.17	14.52	175.25	1,123.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	12.08	0.00	0.00	321.61	4,749,626.00 *gal*
Application event totals		12.08	0.00	0.00	321.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.74	8.32	108.90	747.47	450,000.00 *gal*
Canal	Surface water	10.62	0.00	0.00	282.84	4,177,028.00 *gal*
Application event totals		56.37	8.32	108.90	1,030.32	

Field 2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.17	7.48	98.01	672.73	405,000.00 *gal*
Canal	Surface water	10.17	0.00	0.00	270.83	3,999,689.00 *gal*
Application event totals		51.34	7.48	98.01	943.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.03	8.73	114.35	784.85	472,500.00 *gal*
Canal	Surface water	12.14	0.00	0.00	323.25	4,773,767.00 *gal*
Application event totals		60.17	8.73	114.35	1,108.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.62	0.00	0.00	282.84	4,177,028.00 *gal*
Application event totals		10.62	0.00	0.00	282.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.08	0.00	0.00	188.56	2,784,685.00 *gal*
Application event totals		7.08	0.00	0.00	188.56	

Field 3 - 10/11/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 10/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	83.71	19.45	103.16	0.00	145.00 *ton*
Application event totals		83.71	19.45	103.16	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.49	8.91	131.81	796.90	405,000.00 *gal*
Canal	Surface water	5.75	0.00	0.00	153.00	2,044,262.00 *gal*
Application event totals		41.23	8.91	131.81	949.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.37	14.90	179.86	899.28	292,500.00 *gal*
Canal	Surface water	6.22	0.00	0.00	165.62	2,212,951.00 *gal*
Application event totals		103.59	14.90	179.86	1,064.90	

Field 3 - 06/20/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.37	14.90	179.86	899.28	292,500.00 *gal*
Canal	Surface water	11.26	0.00	0.00	299.70	4,004,398.00 *gal*
Application event totals		108.62	14.90	179.86	1,198.98	

Field 3 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.36	0.00	0.00	249.15	3,329,020.00 *gal*
Application event totals		9.36	0.00	0.00	249.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.50	8.27	108.33	743.54	405,000.00 *gal*
Canal	Surface water	9.78	0.00	0.00	260.26	3,477,483.00 *gal*
Application event totals		55.28	8.27	108.33	1,003.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.45	7.35	96.29	660.92	360,000.00 *gal*
Canal	Surface water	10.66	0.00	0.00	283.92	3,793,623.00 *gal*
Application event totals		51.11	7.35	96.29	944.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.50	8.27	108.33	743.54	405,000.00 *gal*
Canal	Surface water	11.06	0.00	0.00	294.44	3,934,134.00 *gal*
Application event totals		56.56	8.27	108.33	1,037.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.74	0.00	0.00	312.61	4,177,028.00 *gal*
Application event totals		11.74	0.00	0.00	312.61	

Field 3 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.83	0.00	0.00	208.41	2,784,685.00 *gal*
Application event totals		7.83	0.00	0.00	208.41	

Field 4 - 10/11/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	50.14	11.65	61.79	0.00	96.00 *ton*
Application event totals		50.14	11.65	61.79	0.00	

11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.67	8.96	132.51	801.12	450,000.00 *gal*
Canal	Surface water	7.80	0.00	0.00	207.64	3,066,393.00 *gal*
Application event totals		43.47	8.96	132.51	1,008.76	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.32	12.45	150.21	751.05	270,000.00 *gal*
Canal	Surface water	6.69	0.00	0.00	178.10	2,630,143.00 *gal*
Application event totals		88.01	12.45	150.21	929.15	

Field 4 - 06/20/2023: Corn, silage

Field 4 - 06/20/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.87	14.52	175.25	876.23	315,000.00 *gal*
Canal	Surface water	9.30	0.00	0.00	247.72	3,658,342.00 *gal*
Application event totals		104.17	14.52	175.25	1,123.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	12.08	0.00	0.00	321.61	4,749,626.00 *gal*
Application event totals		12.08	0.00	0.00	321.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.74	8.32	108.90	747.47	450,000.00 *gal*
Canal	Surface water	10.62	0.00	0.00	282.84	4,177,028.00 *gal*
Application event totals		56.37	8.32	108.90	1,030.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.17	7.48	98.01	672.73	405,000.00 *gal*
Canal	Surface water	10.17	0.00	0.00	270.83	3,999,689.00 *gal*
Application event totals		51.34	7.48	98.01	943.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.03	8.73	114.35	784.85	472,500.00 *gal*
Canal	Surface water	12.14	0.00	0.00	323.25	4,773,767.00 *gal*
Application event totals		60.17	8.73	114.35	1,108.10	

Field 4 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.62	0.00	0.00	282.84	4,177,028.00 *gal*
Application event totals		10.62	0.00	0.00	282.84	

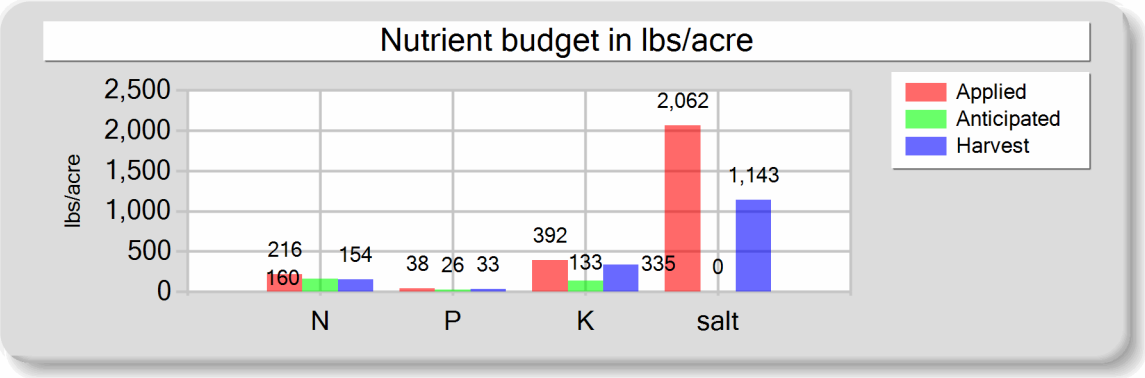
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.08	0.00	0.00	188.56	2,784,685.00 *gal*
Application event totals		7.08	0.00	0.00	188.56	

B. NUTRIENT BUDGET

Field 1 - 10/11/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	61.65	14.32	75.98	0.00
Process wastewater	134.31	24.10	315.62	1,718.61
Fresh water	12.89	0.00	0.00	343.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.85	38.43	391.60	2,061.82
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.19	33.43	335.34	1,142.96
Nutrient balance	61.66	5.00	56.26	918.86
Applied to removed ratio	1.40	1.15	1.17	1.80

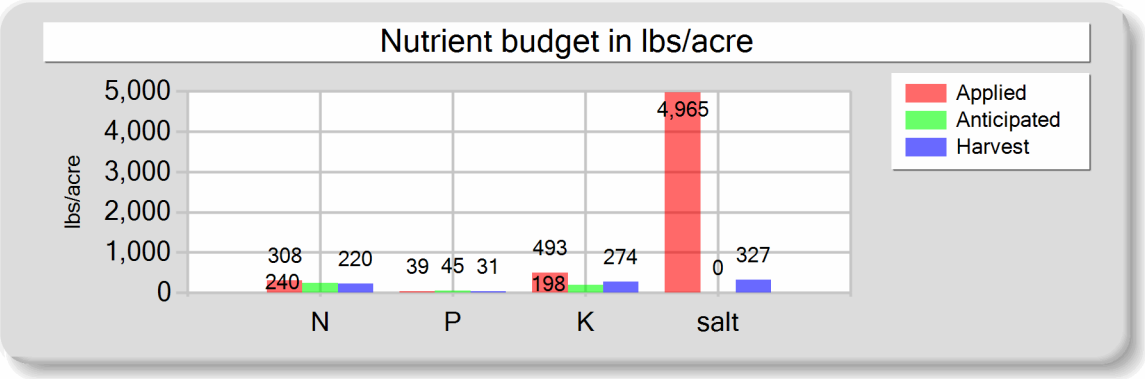
Fresh water applied	
6,999,409.00	*gallons*
257.76	*acre-inches*
8.89	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/20/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.98	38.81	492.79	3,044.47
Fresh water	72.13	0.00	0.00	1,920.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	308.11	38.81	492.79	4,964.82
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	220.18	30.96	273.50	326.83
Nutrient balance	87.93	7.84	219.28	4,637.99
Applied to removed ratio	1.40	1.25	1.80	15.19

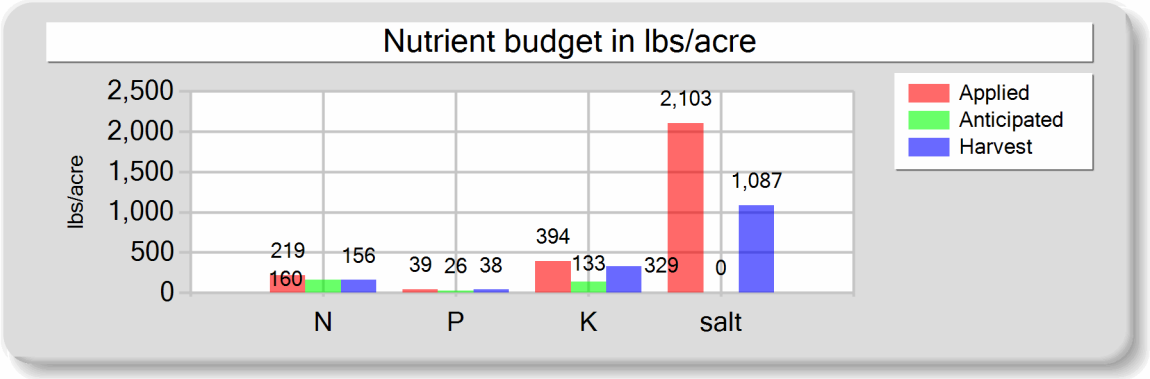
Fresh water applied	
39,163,525.00	*gallons*
1,442.26	*acre-inches*
49.73	*inches/acre*

Process wastewater applied	
2,227,500.00	*gallons*
82.03	*acre-inches*
2.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/11/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	64.77	15.05	79.82	0.00
Process wastewater	132.33	23.93	314.38	1,717.40
Fresh water	14.49	0.00	0.00	385.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.59	38.97	394.20	2,103.13
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	156.01	37.82	329.04	1,087.35
Nutrient balance	62.58	1.15	65.15	1,015.78
Applied to removed ratio	1.40	1.03	1.20	1.93

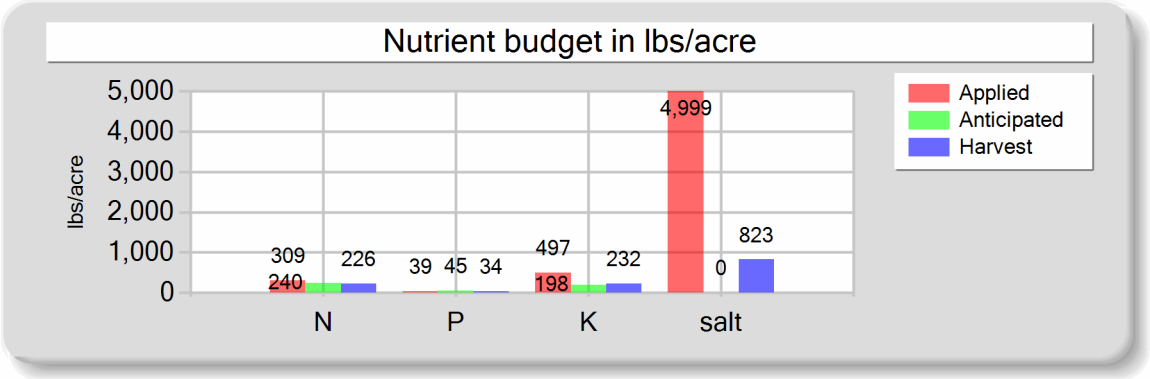
Fresh water applied	
5,696,536.00	*gallons*
209.78	*acre-inches*
9.99	*inches/acre*

Process wastewater applied	
787,500.00	*gallons*
29.00	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/20/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	229.81	39.05	496.51	3,081.27
Fresh water	72.02	0.00	0.00	1,917.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	308.84	39.05	496.51	4,998.94
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	226.42	34.31	231.57	823.36
Nutrient balance	82.41	4.74	264.94	4,175.58
Applied to removed ratio	1.36	1.14	2.14	6.07

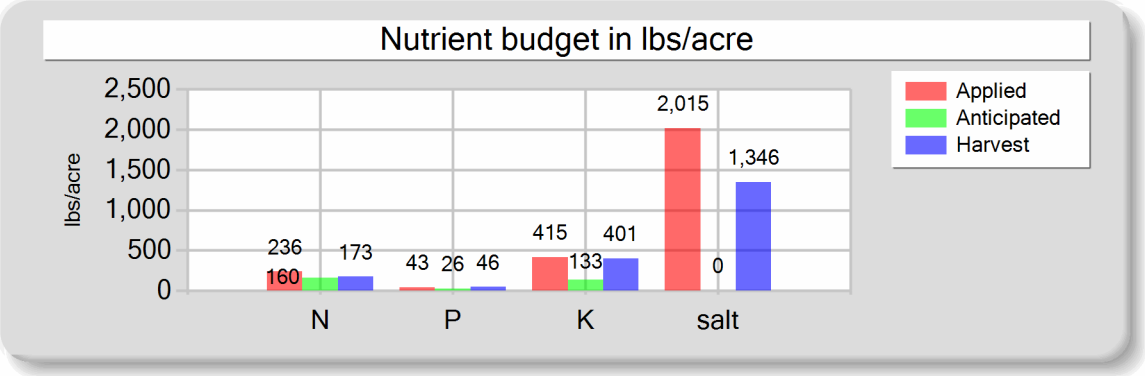
Fresh water applied	
28,320,165.00	*gallons*
1,042.94	*acre-inches*
49.66	*inches/acre*

Process wastewater applied	
1,642,500.00	*gallons*
60.49	*acre-inches*
2.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/11/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	83.71	19.45	103.16	0.00
Process wastewater	132.85	23.81	311.67	1,696.19
Fresh water	11.97	0.00	0.00	318.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.53	43.26	414.83	2,014.80
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	173.08	45.94	400.64	1,346.14
Nutrient balance	62.46	-2.68	14.19	668.66
Applied to removed ratio	1.36	0.94	1.04	1.50

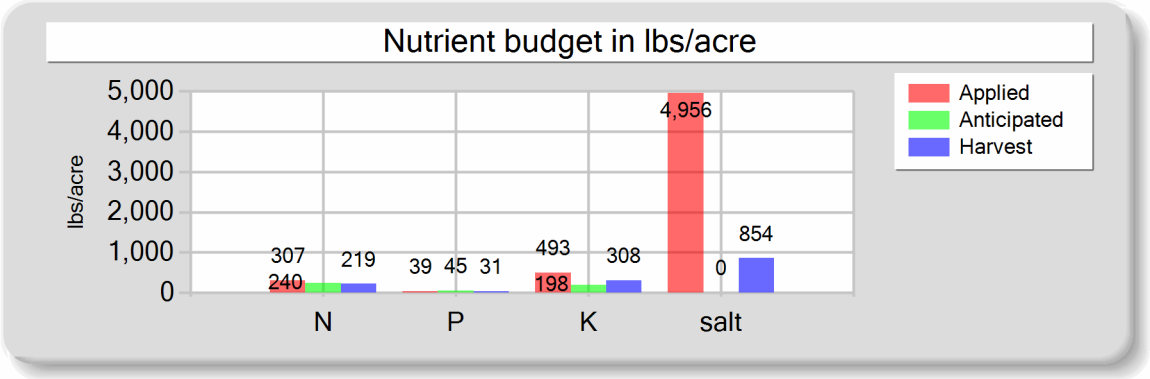
Fresh water applied	
4,257,213.00	*gallons*
156.78	*acre-inches*
8.25	*inches/acre*

Process wastewater applied	
697,500.00	*gallons*
25.69	*acre-inches*
1.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/20/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.82	38.80	492.81	3,047.29
Fresh water	71.68	0.00	0.00	1,908.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	307.50	38.80	492.81	4,955.77
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	219.13	30.61	307.74	853.95
Nutrient balance	88.37	8.18	185.06	4,101.82
Applied to removed ratio	1.40	1.27	1.60	5.80

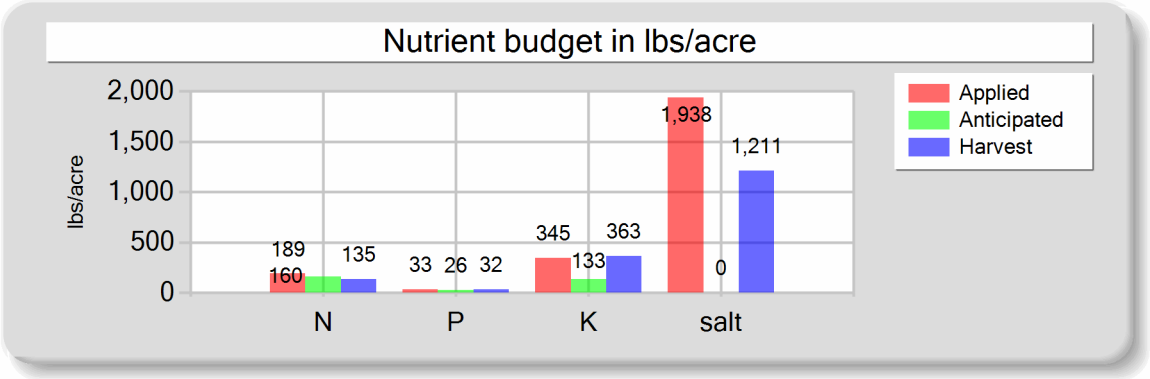
Fresh water applied	
25,500,371.00	*gallons*
939.09	*acre-inches*
49.43	*inches/acre*

Process wastewater applied	
1,462,500.00	*gallons*
53.86	*acre-inches*
2.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 10/11/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	50.14	11.65	61.79	0.00
Process wastewater	116.99	21.40	282.72	1,552.17
Fresh water	14.49	0.00	0.00	385.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	188.62	33.05	344.51	1,937.90
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	134.87	32.49	363.26	1,210.88
Nutrient balance	53.75	0.57	-18.75	727.02
Applied to removed ratio	1.40	1.02	0.95	1.60

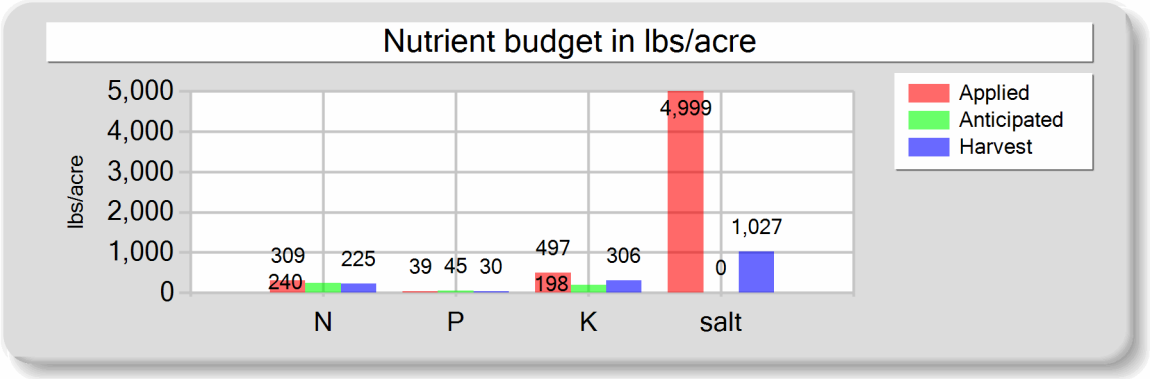
Fresh water applied	
5,696,536.00	*gallons*
209.78	*acre-inches*
9.99	*inches/acre*

Process wastewater applied	
720,000.00	*gallons*
26.52	*acre-inches*
1.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/20/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	229.81	39.05	496.51	3,081.27
Fresh water	72.02	0.00	0.00	1,917.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	308.84	39.05	496.51	4,998.94
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	224.65	30.49	306.49	1,026.99
Nutrient balance	84.18	8.56	190.02	3,971.95
Applied to removed ratio	1.37	1.28	1.62	4.87

Fresh water applied	
28,320,165.00	*gallons*
1,042.94	*acre-inches*
49.66	*inches/acre*

Process wastewater applied	
1,642,500.00	*gallons*
60.49	*acre-inches*
2.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Manure - 2022

Sample and source description: Corral Manure - 2022
Sample date: 10/20/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 44.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,900.00	2,300.00	12,200.00							0.00
DL	100.00	100.00	100.00							1.00

Corral Manure

Sample and source description: Corral Manure
Sample date: 05/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 56.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,300.00	10,800.00	39,100.00							0.00
DL	100.00	100.00	100.00							1.00

Corral Manure

Sample and source description: Corral Manure
Sample date: 10/20/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 46.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,800.00	5,800.00	23,900.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q3 - 2022

Sample and source description: Lagoon Q3 - 2022

Sample date: 08/31/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	197.00	180.00	0.00	2.50	50.10	741.00								7,450.00	4,480
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,670.00	478.00	0.00	1.70	298.00	1,500.00								11,800.00	5,300
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	757.00	438.00	0.00	0.90	116.00	1,400.00								11,900.00	7,000
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	244.00	174.00	0.00	11.80	46.50	609.00								6,540.00	4,180
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,050.00	413.00	0.00	1.10	406.00	897.00								8,420.00	3,250
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

TID Canal

Sample description: TID Canal

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.40	0.00	6.40								284.00	
DL	0.01	0.01	0.01								10.00	

Domestic Well

DW#1

Sample description: DW#1

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.30	0.00	0.30								1,780.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW#2

Sample description: DW#2

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.60	0.00	1.60								2,640.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 10/11/2022: Oats, silage-soft dough

Oats Field 1

Sample and source description: Oats Field 1

Sample date: 05/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,100.00	31,100.00		10.60
DL	0.01	0.01	0.01		1.00

Field 1 - 06/20/2023: Corn, silage

Field 1 - 06/20/2023: Corn, silage

Field 1 Corn

Sample and source description: Field 1 Corn

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	1,800.00	15,900.00		1.90
DL	0.01	0.01	0.01		1.00

Field 2 - 10/11/2022: Oats, silage-soft dough

Oats Field 2

Sample and source description: Oats Field 2

Sample date: 05/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	4,000.00	34,800.00		11.50
DL	0.01	0.01	0.01		1.00

Field 2 - 06/20/2023: Corn, silage

Field 2 Corn

Sample and source description: Field 2 Corn

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,000.00	13,500.00		4.80
DL	0.01	0.01	0.01		1.00

Field 3 - 10/11/2022: Oats, silage-soft dough

Oats Field 3

Sample and source description: Oats Field 3

Sample date: 05/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	4,300.00	37,500.00		12.60
DL	0.01	0.01	0.01		1.00

Field 3 - 06/20/2023: Corn, silage

Field 3 Corn

Sample and source description: Field 3 Corn

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	1,900.00	19,100.00		5.30
DL	0.01	0.01	0.01		1.00

Field 4 - 10/11/2022: Oats, silage-soft dough

Oats Field 4

Sample and source description: Oats Field 4

Sample date: 05/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	3,300.00	36,900.00		12.30
DL	0.01	0.01	0.01		1.00

Field 4 - 06/20/2023: Corn, silage

Field 4 Corn

Sample and source description: Field 4 Corn

Sample date: 10/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	1,900.00	19,100.00		6.40
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

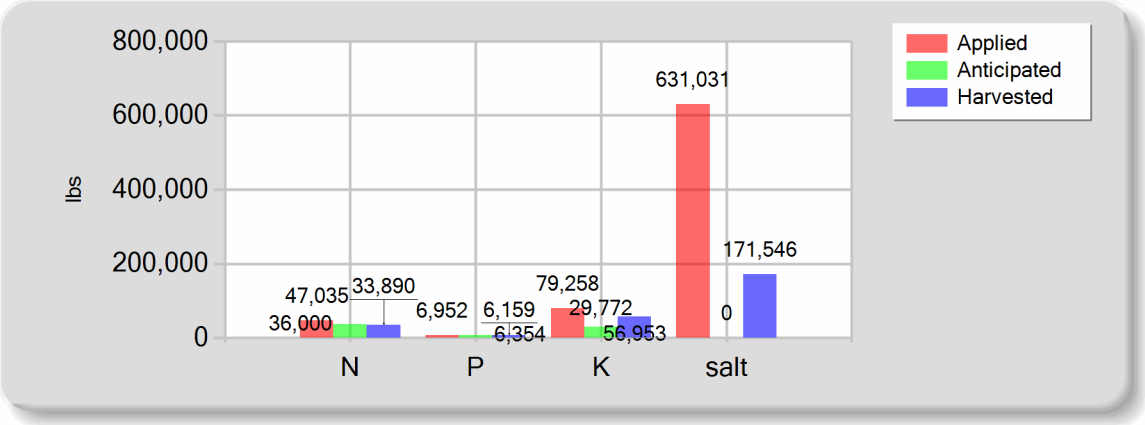
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

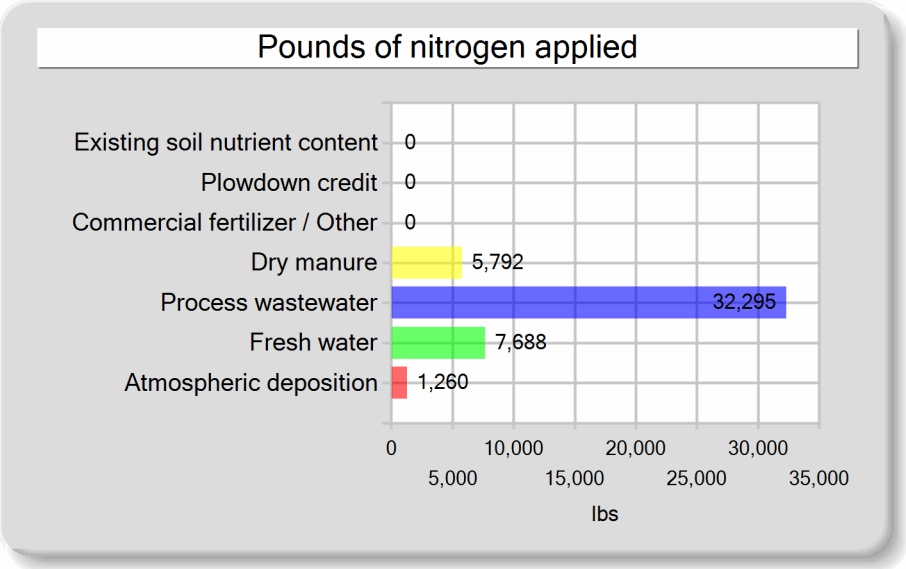
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

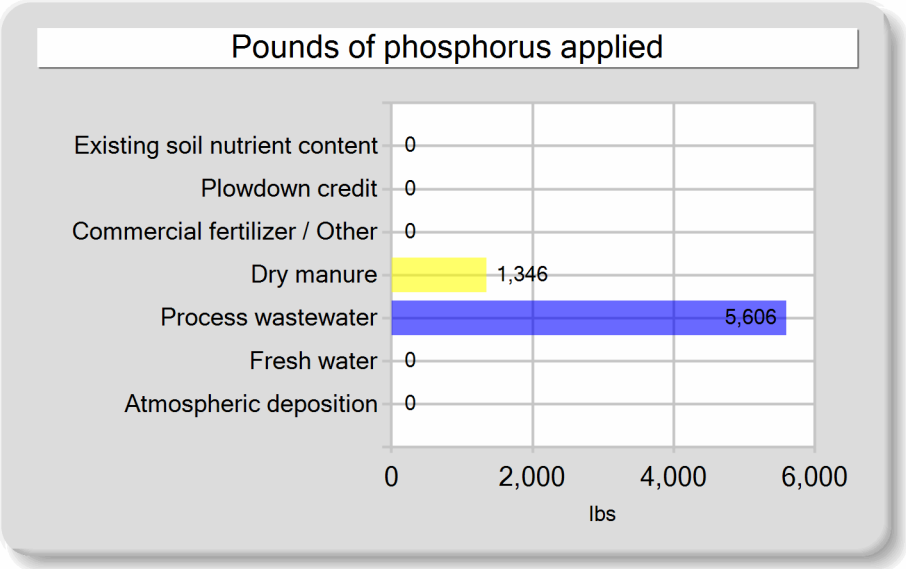
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	5,791.74	1,345.56	7,137.29	0.00
Process wastewater	32,294.91	5,606.06	72,121.08	426,329.78
Fresh water	7,688.29	0.00	0.00	204,700.75
Atmospheric deposition	1,260.00	0.00	0.00	0.00
Total nutrients applied	47,034.93	6,951.62	79,258.37	631,030.52
Anticipated crop nutrient removal	36,000.00	6,354.00	29,772.00	0.00
Actual crop nutrient removal	33,889.74	6,158.96	56,953.45	171,545.76
Nutrient balance	13,145.20	792.66	22,304.93	459,484.77
Applied to removed ratio	1.39	1.13	1.39	3.68

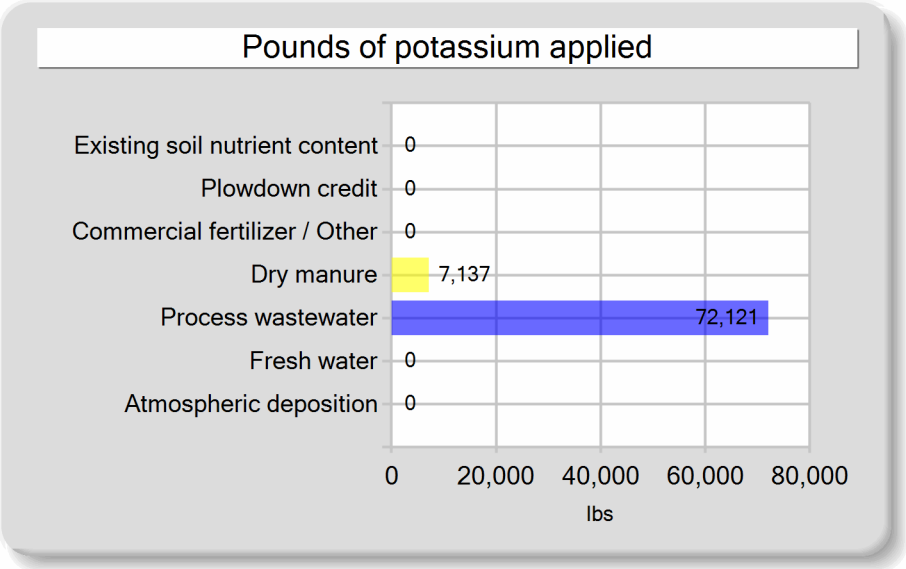
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

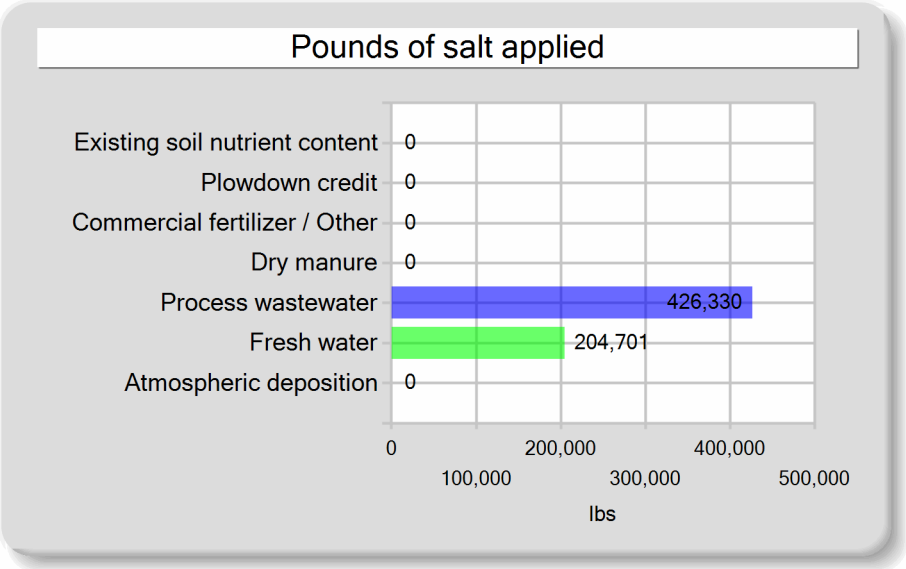


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Jose Soares

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joe & Teresa Soares	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 27, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Jose Soares

Name of Dairy Facility: Joe Soares Dairy #4

Facility Address:

Number and Street	City	County	Zip Code
9207 American Ave	Hilmar	Stanislaus	95324

Contact Person Name and Phone Number:

Name	Phone Number
Jose Soares	(209) 652-2990

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Jose Soares Dairy Pipeline

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 189	Delhi	CA	95315

Contact Person:

Name	Phone Number
Jose Soares	(209) 652-2990

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Art Cabral	(209) 634-8042

Address	City	State	Zip Code
8099 Central AVE	Hilmar	CA	95324

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	Hilmar	95324

Street and nearest cross street (if no address)	County
Central Ave	Merced

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 05/08/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 5,850,000 gallons

Method used to determine volume of process wastewater:

65 hrs x 1500 gpm

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Jose Soares 5/8/23

Operator Signature Date

Jose Soares 5/8/23

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

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2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Jose Soares

Name of Dairy Facility: Joe Soares Dairy #4

Facility Address:

Number and Street	City	County	Zip Code
9207 American Ave	Hilmar	Stanislaus	95324

Contact Person Name and Phone Number:

Name	Phone Number
Jose Soares	(209) 652-2990

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Cardoso Farms Services Inc

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 60	Hilmar	CA	95324

Contact Person:

Name	Phone Number
Arlindo Cardoso	(209) 564-6329

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Art Cabral	(209) 634-8042

Address	City	State	Zip Code
8099 Central AVE	Hilmar	CA	95324

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	Hilmar	95324

Street and nearest cross street (if no address)	County
Central Ave	Merced

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 05/16/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 3,820.00 tons

Manure Solids Content: 43.8 %

Method used to determine amount of manure:

tons per load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Jose Soares 5/16/23

Operator Signature Date

Arlindo Cardoso 5/16/23

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

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2) If there are multiple destinations, complete a separate form for each destination.
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4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Jose Soares

Name of Dairy Facility: Joe Soares Dairy #4

Facility Address:

Number and Street	City	County	Zip Code
9207 American Ave	Hilmar	Stanislaus	95324

Contact Person Name and Phone Number:

Name	Phone Number
Jose Soares	(209) 652-2990

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Jose Soares Dairy Pipeline

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 189	Delhi	CA	95315

Contact Person:

Name	Phone Number
Jose Soares	(209) 652-2990

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Art Cabral	(209) 634-8042

Address	City	State	Zip Code
8099 Central AVE	Hilmar	CA	95324

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	Hilmar	95324

Street and nearest cross street (if no address)	County
Central Ave	Merced

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 06/20/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 5,850,000 gallons

Method used to determine volume of process wastewater:

65 hours x 1500 gpm

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Jose Soares — 6/20/23

Operator Signature — Date

Jose Soares — 6/20/23

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

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2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Jose Soares

Name of Dairy Facility: Joe Soares Dairy #4

Facility Address:

Number and Street	City	County	Zip Code
9207 American Ave	Hilmar	Stanislaus	95324

Contact Person Name and Phone Number:

Name	Phone Number
Jose Soares	(209) 652-2990

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Jose Soares Dairy Pipeline

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 189	Delhi	CA	95315

Contact Person:

Name	Phone Number
Jose Soares	(209) 652-2990

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Art Cabral	(209) 634-8042

Address	City	State	Zip Code
8099 Central AVE	Hilmar	CA	95324

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	Hilmar	95324

Street and nearest cross street (if no address)	County
Central Ave	Merced

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 08/02/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 6,300,000 gallons

Method used to determine volume of process wastewater:

70 hrs x 1500 gpm

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Jose Soares 8/2/23

Operator Signature Date

Jose Soares 8/2/23

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Jose Soares

Name of Dairy Facility: Joe Soares Dairy #4

Facility Address:

Number and Street	City	County	Zip Code
9207 American Ave	Hilmar	Stanislaus	95324

Contact Person Name and Phone Number:

Name	Phone Number
Jose Soares	(209) 652-2990

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Five Star Custom Spreading

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 488	Gustine	CA	95322

Contact Person:

Name	Phone Number
Eric Rocha	(209) 678-2917

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Art Cabral	(209) 634-8042

Address	City	State	Zip Code
8099 Central AVE	Hilmar	CA	95324

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	Hilmar	95324

Street and nearest cross street (if no address)	County
Central Ave	Merced

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 10/18/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 4,840.00 tons

Manure Solids Content: 53.2 %

Method used to determine amount of manure:

tons per load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Jose Soares — 10/18/23

Operator Signature — Date

Eric Rocha — 10/18/23

Hauler Signature — Date

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Soares & Sons Dairy #4
PO Box 189
Delhi CA 95315
00-0022846 08

E-mail: jtsoaresdairy@aol.com
Copy To: cas.labs@yahoo.com

Lab No.: 23L0714
Sampled By: M Dekeyrel
Requested By: Manuela Avila
Submitted Date: 12/12/23
Reported Date: 12/18/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW#1	12/11/23	15:30	1780	1.78	0.3	ND		7.8
2	DW#2	12/11/23	15:35	2640	2.64	1.6	ND		7.8

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Soares & Sons Dairy #4
PO Box 189
Delhi CA 95315
00-0022846 08

E-mail: jtsoaresdairy@aol.com
Copy To: cas.labs@yahoo.com

Lab No.: 23L0714
Sampled By: M Dekeyrel
Requested By: Manuela Avila
Submitted Date: 12/12/23
Reported Date: 12/18/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW#1	12/11/23	15:30	**1780**	**1.78**	0.3	0.82	0.07
2	DW#2	12/11/23	15:35	**2640**	**2.64**	1.6	4.37	0.36

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Soares & Sons Dairy #4
PO Box 189
Delhi, CA 95315

Account# 00-0022846
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/12/2023 15:52
Reported: 12/18/2023 13:58

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0714-01	DW#1	Well Water	M Dekeyrel		12/11/2023 15:30
23L0714-02	DW#2	Well Water	M Dekeyrel		12/11/2023 15:35

Default Cooler Temperature on Receipt °C: -1.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Soares & Sons Dairy #4
PO Box 189
Delhi, CA 95315

Account# 00-0022846
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/12/2023 15:52
Reported: 12/18/2023 13:58

Sample Results

Sample: DW#1
23L0714-01 (Water)

Sampled: 12/11/2023 15:30
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.78**	mmhos/cm	0.01	1		12/13/23 17:56	SM 2510 B		BEL0498
Electrical Conductivity umhos	**1780**	umhos/cm	10.0	1		12/13/23 17:56	SM 2510 B		BEL0498
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 15:30	Field		BEL0393
Nitrate Nitrogen as NO3N	**0.3**	mg/L	0.1	1	10	12/13/23 13:23	EPA 300.0		BEL0351
Temperature	**25.0**	units	0.0	1		12/13/23 17:56	SM 4500-H+	H	BEL0498
pH	**7.8**	units	1.0	1		12/13/23 17:56	SM 4500-H+	H	BEL0498

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LABORATORY INC

Soares & Sons Dairy #4
PO Box 189
Delhi, CA 95315

Account# 00-0022846
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/12/2023 15:52
Reported: 12/18/2023 13:58

Sample Results
(Continued)

Sample: DW#2
23L0714-02 (Water)

Sampled: 12/11/2023 15:35
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.64**	mmhos/cm	0.01	1		12/13/23 17:57	SM 2510 B		BEL0498
Electrical Conductivity umhos	**2640**	umhos/cm	10.0	1		12/13/23 17:57	SM 2510 B		BEL0498
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 15:35	Field		BEL0393
Nitrate Nitrogen as NO3N	**1.6**	mg/L	0.1	1	10	12/13/23 13:43	EPA 300.0		BEL0351
Temperature	**25.0**	units	0.0	1		12/13/23 17:57	SM 4500-H+	H	BEL0498
pH	**7.8**	units	1.0	1		12/13/23 17:57	SM 4500-H+	H	BEL0498

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DELLAVALLE™ LABORATORY INC

Soares & Sons Dairy #4
PO Box 189
Delhi, CA 95315

Account# 00-0022846
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/12/2023 15:52
Reported: 12/18/2023 13:58

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0351										
Blank (BEL0351-BLK1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0351-BLK2)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0351-BLK3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0351-BLK4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0351-BLK5)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0351-BS1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		101	90-110		
LCS (BEL0351-BS2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
LCS (BEL0351-BS3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
LCS (BEL0351-BS4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
Duplicate (BEL0351-DUP1)		**Source: 23L0613-01**			Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	1.8		0.1	mg/L		1.8			0.900	10
Duplicate (BEL0351-DUP2)		**Source: 23L0664-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	6.7		0.1	mg/L		6.7			0.0448	10
Duplicate (BEL0351-DUP3)		**Source: 23L0695-01**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.1		0.1	mg/L		4.1			0.441	10
Duplicate (BEL0351-DUP4)		**Source: 23L0718-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	1.3		0.1	mg/L		1.3			1.70	10
Matrix Spike (BEL0351-MS1)		**Source: 23L0613-01**			Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	7.1		0.1	mg/L	5.000	1.8	106	90-110		
Matrix Spike (BEL0351-MS2)		**Source: 23L0664-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	11.7		0.1	mg/L	5.000	6.7	101	90-110		

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DELLAVALLE™ LABORATORY INC

Soares & Sons Dairy #4
PO Box 189
Delhi, CA 95315

Account# 00-0022846
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/12/2023 15:52
Reported: 12/18/2023 13:58

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0351 (Continued)										
Matrix Spike (BEL0351-MS3)			**Source: 23L0695-01**		Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.3		0.1	mg/L	5.000	4.1	105	90-110		
Matrix Spike (BEL0351-MS4)			**Source: 23L0718-02**		Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	6.6		0.1	mg/L	5.000	1.3	105	90-110		
Reference (BEL0351-SRM1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0351-SRM2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0351-SRM3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0351-SRM4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0351-SRM5)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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LABORATORY INC

Soares & Sons Dairy #4
PO Box 189
Delhi, CA 95315

Account# 00-0022846
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/12/2023 15:52
Reported: 12/18/2023 13:58

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0498										
Blank (BEL0498-BLK1)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0498-BLK2)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Blank (BEL0498-BLK3)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0498-DUP1)		**Source: 23L0716-01**			Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	2.02		0.01	mmhos/cm		2.00			0.916	10
pH	7.4		1.0	units		7.4			0.135	10
Electrical Conductivity umhos	2020		10.0	umhos/cm		2000			0.916	10
Duplicate (BEL0498-DUP2)		**Source: 23L0724-02**			Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	2.08		0.01	mmhos/cm		2.04			2.11	10
Electrical Conductivity umhos	2080		10.0	umhos/cm		2040			2.11	10
pH	7.3		1.0	units		7.3			0.274	10
Reference (BEL0498-SRM1)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	436			umhos/cm	426.0		102	90-110		
Reference (BEL0498-SRM2)					Prepared & Analyzed: 12/13/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0498-SRM3)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0498-SRM4)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEL0498-SRM5)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		

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DELLAVALLE™
LABORATORY INC

Soares & Sons Dairy #4
PO Box 189
Delhi, CA 95315

Account# 00-0022846
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/12/2023 15:52
Reported: 12/18/2023 13:58

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0498 (Continued)										
Reference (BEL0498-SRM5)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEL0498-SRM6)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0498-SRM7)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0498-SRM8)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/12/23 15:52 23L0714

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Acct No. / Cons.

Bill To: 22846 | 8

Purchase Order No. _____ Results Needed By _____

Client: **Soares Dairy #4, Jose D.**
Address: P.O. Box 189
City, State, Zip: Delhi, CA 95315
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Manuela Avila

Ranch _____

Date sampled: 12/11/23

Sampled by: MDekeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 2 No. Bottles _____

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

[X] DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES

		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW#1	Sampled From:	12/11/23	330p	Ø	-1.6
2. DW#2	Sampled From:	12/11/23	335p	Ø	-2.0
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MDekeyrel	CASINC	12/11/23 335p	12/12/23 @ 9:30a
Second	JLF	DLV	12-12-23 9:30	12-12-23 2:01
Third	MM	DLI	12/12 15:52	
Fourth				

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping

$ _____ In
$ _____ Out

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	1	1								
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

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SEND OUT